Abhishek Bhattarai

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Senior Software Engineer | Full-Stack Engineer | Technical Lead

Senior Software Engineer with **4+ years** of experience shipping products used by **1M+ monthly learners** and leading a team of developers at Programiz, a platform with **2.5M+ monthly users**. Cut authentication latency by **45%**, reduced deployment cycles from **14 days to 4**, and halved course production time from **12 days to 6**. Now building Turnit at reAlpha, owning features from system design through CI/CD, testing, and production. Full-stack across TypeScript, Node.js, React, and PostgreSQL.

Technical Skills

Languages: TypeScript, JavaScript, Python, SQL
Frontend: React, Next.js, React Native, Tailwind CSS
Backend: Node.js, Express.js, REST APIs, GraphQL, PostgreSQL, MongoDB, MySQL, WebSockets, caching, message queues, background jobs, Kafka
Architecture & Quality: System design, API design, database design, performance tuning, code review, Jest, React Testing Library
Cloud & DevOps: AWS, Docker, CI/CD, release management, Sentry, observability
AI & Data: Vercel AI SDK, LLM APIs (GPT, Gemini), Pinecone, BigQuery

Professional Experience

reAlpha Tech Corp. (Nasdaq: AIRE) May 2026 – Present
Senior Software Engineer Ohio, USA | Remote

- Own end-to-end delivery of Turnit, a real estate turnover management platform used by property managers, vendors, and operations teams, spanning React, Next.js, React Native, Node.js, and TypeScript.
- Design and build GraphQL APIs, backend services, and PostgreSQL data models, including client-server integrations and module boundaries.
- Own and maintain build, test, and deployment pipelines (CI/CD) and release processes using AWS and Docker.
- Write and maintain automated tests with Jest and React Testing Library to protect critical user flows and catch regressions before release.
- Delivered an AI-powered product chatbot with the Vercel AI SDK and Pinecone, giving users retrieval-based access to product knowledge.
- Contribute to technical design discussions, code reviews, and technical documentation.

Programiz August 2025 – April 2026
Senior Software Engineer Lalitpur, Nepal

- Led a team of developers: assigned work, reviewed code, mentored engineers, and tracked and reported progress to project management.
- Led end-to-end engineering of the Python, C, and DSA Code Visualizer Suite, serving 100K+ monthly learners.
- Built the execution pipeline connecting TypeScript applications to Python services, converting debugger state into structured data for step-by-step visualization.
- Reduced authentication latency by 45% by moving non-critical work to background jobs outside the blocking request path.
- Designed an AI-assisted content workflow with GPT, Gemini, and Slack APIs, cutting course production time from 12 days to 6.
- Built a server-side tracking pipeline with server-side Google Tag Manager and Meta Conversions API, improving tracking accuracy by 50%.

Programiz July 2022 – August 2025
Software Engineer Lalitpur, Nepal

- Developed frontend and backend services for a browser-based programming environment supporting eight languages and 1M+ monthly learners.
- Built and maintained REST APIs for lessons and quizzes handling millions of requests per month.

- Engineered Python and Node.js workflow automation that reduced deployment cycles from 14 days to 4.
- Implemented CI/CD and deployment automation to improve consistency and reliability across environments.
- Built an internal administration dashboard and feature-flag controls to support experimentation and operational visibility.

Selected Projects

Grievance Desk Live demo | GitHub

TypeScript, React, Express, PostgreSQL, Neon, Vercel

- Built a complaint review desk that stages CSV imports, counts five working days in the Asia/Kathmandu time zone, and records each supervisor decision in the same database transaction as its follow-up, so every import, deadline, and decision stays explainable.
- Enforced per-user access so a signed-in email sees only its own cases; deployed Express on Vercel and PostgreSQL on Neon with migrations applied from CI.

Browser-Based IDE (Programiz Playground) Live demo

React, Next.js, Node.js, REST APIs, real-time code execution

- Primary developer of a multi-language online IDE supporting 8 programming languages with real-time, isolated code execution; used by 1M+ monthly learners.
- Built backend APIs and frontend components, and contributed to deployment and reliability improvements, powering examples, lessons, and project-based learning.
- Platform context (company-wide, per Programiz media kit): programiz.com reaches 2.5M+ unique users and 9.5M+ page views per month.

Code Visualizer Suite (Programiz) Live demo

Python, TypeScript, React, Node.js, REST APIs

- Led the build of Python, C, and DSA visualizers with line-by-line execution, variable tracking, and recursion views for 100K+ monthly active learners.
- Developed execution services and data pipelines that turn runtime debugger state into structured data for the frontend.

AI-Assisted Content Workflow (Programiz)

OpenAI, Gemini, Slack API, prompt engineering

- Built a Slack-integrated, multi-stage GPT and Gemini workflow that turns raw documents into structured courses, reducing production time from 12 days to 6.

Education

Coventry University September 2019 – September 2022

BSc (Hons) Cyber Security and Ethical Hacking | GPA: 3.6+

Additional Experience

Resource Person / Trainer, Edulift Organization December 2025

Trained 18 government schoolteachers on interactive lessons, quizzes, and Google Classroom.

Applied Data Science Labs, WorldQuant University August 2023